Katherine Braught

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Education

University of Illinois Urbana–Champaign — **Area: Computers and Education**
Ph.D. in Computer Science, Advisor: Jeff Erickson and Colleen Lewis — *Aug 2021–Present*

University of Illinois Urbana–Champaign — **Area: Formal Methods for Verification**
M.S. in Computer Science, Advisor: Sayan Mitra — *Aug 2021–Aug 2024*

Iowa State University, Ames, Iowa — **Minor: Biology**
B.S. in Computer Science, University Honors — *Aug 2016–Dec 2020*
Summa Cum Laude, Top 2% of LAS College

Research Interests

My research focuses on making students' experiences in upper-level computer science theory courses engaging and supportive. I approach this by (1) investigating the culture and experiences of students in algorithms courses, (2) developing personalized problem sets for these courses, and (3) training teaching assistants to teach problem-solving effectively.

Teaching

Instructor Experience

CS 173: Discrete Structures, UIUC
Instructor — *Summer 2026*
- Designed and delivered four 30-minute lectures per week for 75 students
- Led four 45-minute group problem-solving sessions per week
- Guided one TA through grading assignments and managed in-class help from three undergraduate assistants

University of Illinois Principal's Scholars Program
Instructor — *Fall 2025, Fall 2026*
- Developing curriculum to support computer science exploration for 15 students in grades 6–8
- Developed curriculum to support pre-algebra skills for 26 students in grades 6–8

Illinois Mathematics and Science Academy
Motorola CS+X Summer Campus Instructor — *June 2025*
Taught a 40-hour course to 20 middle school students with a CS + linguistics project. Developed instructional materials that linked Python and core computing concepts to real-world contexts

Teaching Assistant Experience

CS 591: Teaching Assistant Training, UIUC
Course Assistant — *Aug 2024–Present*
- Manage attendance of approximately 300 students
- Grade reflection assignments
- Contribute to improving TA training lectures

CS 374: Intro to Algs and Models of Comp, UIUC
Teaching Assistant and Discussion Section Leader — *Aug 2025–May 2026*
- Led group problem-solving sessions with 100 students twice a week
- Supported undergraduate assistants with grading and graded exams

CS 173: Discrete Structures, UIUC
Teaching Assistant *Aug 2024–May 2025*
- Taught weekly tutorials for approximately 200 students and held office hours
- Graded biweekly proof-based exams

COM S 311: Introduction to Algorithm Analysis and Design, Iowa State University
Teaching Assistant *Aug 2019–Nov 2020*
- Taught a weekly recitation for approximately 25 students and held weekly office hours
- Graded theoretical and application-based algorithm assignments

COM S 327: Advanced Programming Techniques, Iowa State University
Teaching Assistant *Aug 2018–Dec 2018*
- Supported students during weekly office hours
- Graded and debugged C/C++ programming assignments

COM S 207: Careers in Computer Science, Iowa State University
Teaching Assistant *Jan 2018–May 2018*
- Managed course activities and grading for a 135-student course

BIOL 212: Principles of Biology II, Iowa State University
Learning Assistant *Aug 2017–Dec 2017*
- Facilitated discussion of complex biological concepts with small groups of students
- Planned active-learning activities with teaching staff

Guest Lecturing

Teaching and Leadership Eng 590
Teaching Engineering Problem-Solving and Meeting Students Where They AreSpring 2026; Fall 2025; Spring 2024
Lectured to approximately 300 graduate teaching assistants on teaching problem-solving

UIUC's Grad Academy
Using Compassionate Pedagogy as a TA *Fall 2025*
Led an active-learning workshop for approximately 60 students on compassionate pedagogy

UIUC's Grad Academy
Teaching Computer Science Problem-Solving and Meeting Students Where They Are *Spring 2025*
Lectured to approximately 40 computer science graduate teaching assistants on teaching problem-solving

Professional Work Experiences

IBM
Backend Developer: Systems *Jan 2021–Aug 2021*
- Developed software in a proprietary language for IBM i
- Implemented new data type support for databases

Collins Aerospace
Platform Software Engineering Co-op *Jan 2019–July 2019*
- Developed platform-level software in C for touchscreen hardware
- Created and maintained nightly continuous integration testing using Atlassian tools and Tcl

Workiva Inc.
Software Development Intern *May 2018–Dec 2018*
- Developed Java microservices on an Agile software development team
- Maintained existing applications and implemented new features

Research Internship Experience

NASA Langley **Mentor: Alwyn Goodloe, Ph.D.**
Ph.D. Student Researcher *May 2023–Aug 2023*

- Implemented probabilistic monitoring in Copilot
- Developed monitoring examples for safety-critical systems

Air Force Research Lab, Wright-Patterson AFB **Mentor: Kerrianne Hobbs, Ph.D.**
Ph.D. Student Researcher *May 2022–Aug 2022*

- Synthesized a reinforcement learning controller for the Inspection Problem
- Integrated runtime assurance into RL controllers

Carnegie Mellon University, Software Engineering Institute **Mentor: Eunsuk Kang, Ph.D.**
NSF REU Researcher *May 2020–Sept 2020*

- Applied formal methods to reason about erroneous human behavior in medical devices
- Developed a tool to generate erroneous behavior from phenotypical errors

Auburn University, Department of Weed and Soil Sciences **Mentor: Scott McElroy, Ph.D.**
NSF Bioinformatics REU Researcher *May 2017–July 2017*

- Analyzed plant genomes using Biopython and Unix scripting on an HPC cluster
- Mentored Ph.D. students learning Python coding

Research Publications and Manuscripts

More Similar Than Different: Accounting for Gender Differences in Algorithms Course Performance
Katherine Braught, Colleen Lewis, Jeff Erickson
In preparation; intended for submission to ACM ITiCSE 2027.

Novice Difficulties with Graph Problem Reduction in Algorithm Design
Hongxuan Chen, **Katherine Braught**, Geoffrey L. Herman, Jeff Erickson
ACM SIGCSE Virtual, 2026 (Accepted).

Evaluating LLM-Generated Contextualized Algorithm Design Problems
Erica Goodwin*, **Katherine Braught***, Jonathan Liu, Dip Kiran Pradhan Newar, Yael Gertner, Seth Poulsen, Diana Franklin
*ACM ICER, 2026, *Equal contribution.*
DOI: 10.1145/3765964.3811658

An Engineering TA Training Lesson on Problem-Solving: Design and Evaluation
Katherine Braught, Blake Johnson, Yael Gertner
ASEE, 2026
DOI: 10.18260/1-2–58733

Interdisciplinary Engineering Education in a Middle School Computer Science Camp
Kathleen Isenegger, **Katherine Braught**, Soha Huq, Cassandra Armstrong, Sharon Doering-Domanus, Yael Gertner, Evan M. Glazer
ASEE, 2026
DOI: 10.18260/1-2–60181

A TA Training Lesson for Problem-Solving
Katherine Braught, Carl Evans, Blake Johnson, Yael Gertner
ACM SIGCSE, 2026
DOI: 10.1145/3770761.3777214

Too Theoretical and Nowhere Near Interesting (Formal Methods)
Katherine Braught, Yangge Li, Katherine Driggs-Campbell, Sayan Mitra
ACM ITiCSE, 2025
DOI: 10.1145/3724363.3729116

Novice Difficulties in Graph Layering for Algorithm Design
Hongxuan Chen, **Katherine Braught**, Geoffrey L. Herman, Jeff Erickson
ACM SIGCSE, 2025
DOI: 10.1145/3641555.3705221

Creating a Community of Graduate Student Computer Science Education Researchers
Grace Barkhuff, **Katherine Braught**, Emma R. Dodoo, Michael Link, Xinying Hou, Elliot Roe
ACM SIGCSE, 2025
DOI: 10.1145/3641555.3705111

Verse: A Python Library for Reasoning About Multi-agent Hybrid System Scenarios
Yangge Li, Haoqing Zhu, **Katherine Braught**, Keyi Shen, Sayan Mitra
CAV, 2023
DOI: 10.1007/978-3-031-37706-8_18

Checking Phylogenetics Decisiveness in Theory and in Practice
Ghazaleh Parvini, **Katherine Braught**, David Fernández-Baca
IEEE/ACM TCBB, 2022
DOI: 10.1109/TCBB.2021.3128381

Honors and Awards

2027: Graduate Teacher Certificate, Expected April 2027

2025–2026: UIUC FLEX Facilitating Learning Excellence Award

The FLEX award recognizes instructors and teaching assistants who achieve outstanding FLEX evaluation results.

2025–2026: University of Illinois Mavis Future Faculty Fellow

2025: UIUC Academy for Excellence in Engineering Education Recognition of Outstanding Student Contribution to a Strategic Instructional Innovations Program project

Service and Leadership

Professional Service

UIUC
Computer Science Culture and Climate Committee — *2026*

ACM ITiCSE
Reviewer — *2025, 2026*

ACM SIGCSE
Reviewer — *2026*

Outreach

UIUC CS+X Workshops
Lead Volunteer Instructor — *2024–Present*
Work with local middle and high school students to teach CS+X workshops

UIUC Outreach Workshops and Principal's Scholars Program
Scholarship Reviewer — *2024–Present*
Work with local students to review scholarship applications for NCWIT Aspirations in Computing and Common App

UIUC Summer Camps
Volunteer Instructor — *Summer 2024*
Worked with Illinois students for a two-day workshop at a summer engineering camp

Chicago Architecture Center
Girls Build! Day Volunteer — *Mar 2023, Aug 2024*
Led teenage girls through hands-on engineering projects aimed at encouraging interest in computing fields

Personal Service

Champaign County Clerk Office
Election Judge — *2024–Present*
Administered election procedures and supported a fair, free, and accessible voting process in Champaign County

Junior League of Champaign-Urbana
Member *2023–2025*
Partnered with local nonprofits and coordinated service projects

Champaign Park District
Special Events Volunteer *2022–2024*
Supported large community events and resident engagement